Khashayar Gatmiry

Berkeley, CA | gatmiry@berkeley.edu | gatmiry.github.io | LinkedIn | Google Scholar

Education

UC Berkeley, Simons Institute: Postdoctoral Researcher — August 2025 - Present
- **Advisors:** Peter Bartlett, Emmanuel Abbe

Apple, MLR team: Part-time Researcher — August 2025 - Present
- **Advisors:** Samy Bengio, Emmanuel Abbe

Massachusetts Institute of Technology: PhD in Computer Science — Sept 2019 – August 2025
- **Advisors:** Stefanie Jegelka, Jonathan Adam Kelner

Sharif University of Technology, B.S. in Computer Engineering & Mathematics — Aug 2014, June 2019
- **Mentors:** Manuel Gomez Rodriguez, Abolfazl Motahari

Research Interests

LLM Reasoning, Reinforcement learaning and LLM post-training, Interpretability, LLM Safety, Optimization for deep learning and LLMs

Selected Publications

- **RLTL;DR: Self-improvement by Internalizing Self-generated Feedback**

  Michael Kirchhof, Eleonora Gualdoni, Andrew Szot, ***Khashayar Gatmiry***, Aryo Lotfi, Omar Attia, Abbas Kazerouni, Sanjoy Chowdhury, Alexander T Toshev,

  PREPRINT (submitted to ICLR 2027)

- **Online Dense Reinforcement Learning with Planning-Grounding Agent for UI Navigation on Live iOS Apps**

  Peitong Duan, ***Khashayar Gatmiry***, Jeffrey Nichols, Samy Bengio, Emmanuel Abbe,

  PREPRINT (submitted to ICLR 2027)

- **Learn Your Own Thoughts: Abstract Token Curriculum**

  Khashayar Gatmiry, Parsa Mirtaheri, Avrajit Ghosh, Jonathan A. Kelner, Holden Lee, Jason D. Lee, Nika Haghtalab, Mikhail Belkin, Emmanuel Abbe, Peter Bartlett,

  PREPRINT (submitted to ICLR 2027)

- **Transformers Learn Two Competing Sorting Algorithms, and Only One Length-Generalizes**

  Khashayar Gatmiry, Nathan W. Henry, Chloe H. Su, Shan Chen, Melanie Weber, Sham Kakade, Emmanuel Abbe, Samy Bengio, Jonathan A. Kelner, Peter Bartlett,

  PREPRINT (submitted to ICLR 2027)

- **Blackwell-Guided Logit Search (BLS): Optimizing Multi-Turn Jailbreaks with Subliminal Steering**

  Khashayar Gatmiry, Timothy Ng, Arhaan Garg, Avrajit Ghosh, Emmanuel Abbe, Peter Bartlett, Bin Yu,

  PREPRINT (submitted to ICLR 2027)

- **High-accuracy and dimension-free sampling with diffusions**

  Khashayar Gatmiry, Sitan Chen, Adil Salim,

  ICML, 2026 (oral)

- **Can Looped Transformers Learn to Implement Multi-step Gradient Descent for In-context Learning?**
  Khashayar Gatmiry, Nikunj Saunshi, Sashank J. Reddi, Stefanie Jegelka, Sanjiv Kumar

ICML, 2024

- **The Inductive Bias of Flatness Regularization for Deep Matrix Factorization**
  Khashayar Gatmiry, Zhiyuan Li, Ching-Yao Chuang, Sashank J. Reddi, Tengyu Ma, Stefanie Jegelka
  NEURIPS, 2023

- **What does guidance do? a fine-grained analysis in a simple setting**
  Muthu Chidambaram, ***Khashayar Gatmiry***, Sitan Chen, Holden Lee, Jianfeng Lu,
  NEURIPS, 2024

Industry Experience

Research Intern, **Microsoft – Redmond, WA** June 2024 – Aug 2024

- *Mentors:* Ronen Eldan, Adil Salim, Yi Zhang
- Worked on generating paragraph semantic embeddings with small language models (SLMs), based on pytorch

Research intern, **Google – New York** Oct 2022 – Jan 2023

- *Mentors:* Sashank J. Reddi, Sobhan Miryoosefi
- Trained language models using novel gradient estimates, authored 4 papers in collaboration with my group

Research Visiting Experience

Visiting Researcher, Yale Computer Science – New Heaven, CT July 2023 – Aug 2023

- **Mentor:** Andre Wibisono
- Worked on Hamiltonian Monte Carlo and Gibbs sampler

Visiting Researcher, Georgia Tech College of Computing – Atlanta, GA Aug 2021 – Sep 2021

- **Mentor:** Santosh S. Vempala
- worked on proving isoperimetric inequalities on manifolds, KLS conjecture, and sampling from polytopes
  Authored 2 publications from projects related to my visit

Research Intern, Max Plank Institute for Software Systems – Kaiserslautern, GER June 2018 – Aug 2018

- **Mentor:** Manuel Gomez Rodriguez
- worked on the network visibility problem and submodularity

Review Service

Reviewer for Neural Information Processing System (**NEURIPS**), Conference on Learning Representations (**ICLR**), Fundamentals of Computer Science (**FOCS**), Symposium on Discrete Algorithms (**SODA**), Journal of Machine Learning Research (**JMLR**), Theoretical Computer Science (**TCS**)

Teaching Service

Teaching Assistant for Introduction to Probability and Statistics (18.05), MIT Spring 2024
Teaching Assistant for Computation Theory and Automata, Sharif university Spring 2018
Teaching Assistant for Artificial Intelligence, Sharif university Spring 2018
Teaching Assistant for Probability and Statistics, Sharif university Fall 2017
Teaching Assistant for Discrete Structures, Sharif university Fall 2016
Teaching Math Olympiad in high schools during my undergraduate studies

Awards

Fellowship from the Institute of Science and Technology Austria (**IST**) Summer 2018
Outstanding Student Award from Sharif University of Technology 2018
Recipient of the grant for exceptional talents *National Elites Foundation of Iran* undergraduate studies
Gold Medal, 27th Iranian National Olmypiad in Mathematics 2013
Ranked 4th, International Tournament of Young Mathematicians (**ITYM**), Germany 2014
Silver Medal, International Mathematics Competition (**IMC**), Taiwan 2012

Bronze Medal, International Mathematics Competition (**IMC**), Indonesia 2011
Gold Medal, Asian Inter-cities Teenagers Mathematics Olympiad (**AITMO**), Nepal 2011

Publications

Learn Your Own Thoughts: Abstract Token Curriculum

Khashayar Gatmiry, Parsa Mirtaheri, Avrajit Ghosh, Jonathan A. Kelner, Holden Lee, Jason D. Lee, Nika Haghtalab, Mikhail Belkin, Emmanuel Abbe, Peter Bartlett,

PREPRINT (submitted to ICLR 2027)

Transformers Learn Two Competing Sorting Algorithms, and Only One Length-Generalizes

Khashayar Gatmiry, ***Nathan W. Henry***, Chloe H. Su, Shan Chen, Melanie Weber, Sham Kakade, Emmanuel Abbe, Samy Bengio, Jonathan A. Kelner, Peter Bartlett,

PREPRINT (submitted to ICLR 2027)

RLTL;DR: Self-improvement by Internalizing Self-generated Feedback

Michael Kirchhof, Eleonora Gualdoni, Andrew Szot, ***Khashayar Gatmiry***, Aryo Lotfi, Omar Attia, Abbas Kazerouni, Sanjoy Chowdhury, Alexander T Toshev

PREPRINT (submitted to ICLR 2027)

Online Dense Reinforcement Learning with Planning-Grounding Agent for UI Navigation on Live IOS Apps

Peitong Duan, ***Khashayar Gatmiry***, Jeffrey Nichols, Samy Bengio, Emmanuel Abbe

PREPRINT, 2026

Blackwell-Guided Logit Search (BLS): Optimizing Multi-Turn Jailbreaks with Subliminal Steering

Khashayar Gatmiry, Timothy Ng, Arhaan Garg, Avrajit Ghosh, Emmanuel Abbe, Peter Bartlett, Bin Yu

PREPRINT (submitted to ICLR 2027)

High-accuracy and dimension-free sampling with diffusions

Khashayar Gatmiry, Sitan Chen, Adil Salim,

ICML, 2026

Learning mixtures of gaussians using diffusion models

Khashayar Gatmiry, Jonathan A. Kelner, Holden Lee,

COLT, 2025

On the Role of Depth and Looping for In-Context Learning with Task Diversity

Khashayar Gatmiry, Nikunj Saunshi, Sashank J. Reddi, Stefanie Jegelka, Sanjiv Kumar,

PREPRINT, 2024

Can Looped Transformers Learn to Implement Multi-step Gradient Descent for In-context Learning?

Khashayar Gatmiry, Nikunj Saunshi, Sashank J. Reddi, Stefanie Jegelka, Sanjiv Kumar,

ICML, 2024

Simplicity Bias via Global Convergence of Sharpness Minimization

Khashayar Gatmiry, Zhiyuan Li, Sashank J. Reddi, Stefanie Jegelka

ICML, 2024

What does guidance do? A fine-grained analysis in a simple setting

Muthu Chidambaram, ***Khashayar Gatmiry***, Sitan Chen, Holden Lee, Jianfeng Lu,

NEURIPS, 2024

Adversarial Online Learning with Temporal Feedback Graphs

Khashayar Gatmiry, Jon Schneider, Stefanie Jegelka,

COLT, 2024

Sampling Polytopes with Riemannian HMC: Faster Mixing via the Lewis Weights Barrier

Khashayar Gatmiry, Jonathan A. Kelner, Santosh Vempala,

COLT, 2024

EM for Mixture of Linear Regression with Clustered Data
Amir Reisizadeh, ***Khashayar Gatmiry***, Asuman Ozdaglar,
AISTATS, 2024

Bandit Algorithms for Prophet Inequality and Pandora's Box
Khashayar Gatmiry, Thomas Kesselheim, Sahil Singla, Yifan Wang
SODA, 2024

Rethinking Invariance in In-context Learning
Lizhe Fang, Yifei Wang, ***Khashayar Gatmiry***, Yisen Wang
ICML WORKSHOP, 2024

Projection-free online convex optimization via efficient newton iterations
Khashayar Gatmiry, Zak Mhammedi
NEURIPS, 2023

The Inductive Bias of Flatness Regularization for Deep Matrix Factorization
Khashayar Gatmiry, Zhiyuan Li, Tengyu Ma, Sashank J. Reddi, Stefanie Jegelka, Ching-Yao Chuang,
NEURIPS, 2023

Quasi-newton steps for efficient online exp-concave optimization
Khashayar Gatmiry, Zak Mhammedi
COLT, 2023

Quasi-newton steps for efficient online exp-concave optimization
Khashayar Gatmiry, Zak Mhammedi
COLT, 2023

When does Metropolized Hamiltonian Monte Carlo provably outperform Metropolis-adjusted Langevin algorithm?
Yuansi Chen, ***Khashayar Gatmiry***
PREPRINT, 2023

A unified approach to controlling implicit regularization via mirror descent
Haoyuan Sun, ***Khashayar Gatmiry***, Kwangjun Ahn, Navid Azizan
JMLR, 2023

Optimal algorithms for group distributionally robust optimization and beyond
Tasuku Soma, ***Khashayar Gatmiry***, Stefanie Jegelka
PREPRINT, 2022

On the generalization of learning algorithms that do not converge
Nisha Chandramoorthy, Andreas Loukas, ***Khashayar Gatmiry***, Stefanie Jegelka
NEURIPS, 2022

Adaptive Generalization and Optimization of Three-Layer Neural Networks
Khashayar Gatmiry, Stefanie Jegelka, Jonathan A. Kelner
ICLR, 2022

Convergence of the riemannian langevin algorithm
Khashayar Gatmiry, Santosh S. Vempala
PREPRINT

Testing determinantal point processes
Khashayar Gatmiry, Maryam Aliakbarpour, Stefanie Jegelka

NeuRIPS, 2020

The network visibility problem
Khashayar Gatmiry, Manuel Gomez-Rodriguez
ACM TOIS, 2021